

Task:

1. Set up and configure an Nginx server on an EC2 instance using Amazon Linux, Ubuntu, and Red Hat.
2. Configure the instance with Nginx using user data (a Bash script). In both cases, a default web page must be displayed.

Expected Outcome:

- Nginx should be successfully running on all instances.
- Accessing the public IP in a browser should display the default web page.
- Automation using user data should eliminate the need for manual setup.

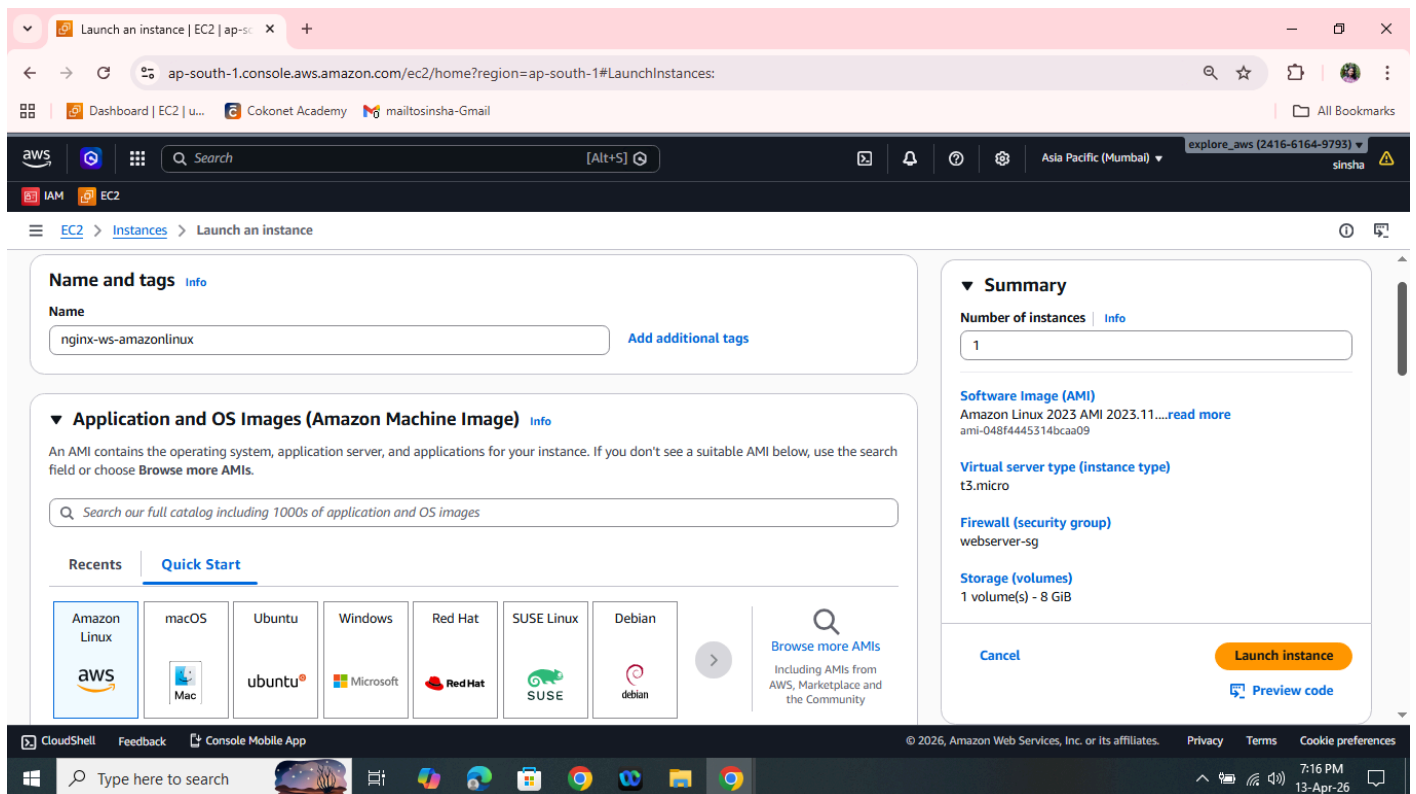
Set up and configure an Nginx server on an EC2 instance using Amazon Linux, Ubuntu, and Red Hat.

PREREQUISITES:

1. Login to AWS account in the preferred region (Eg: Mumbai)
2. Create a Key pair in selected region
3. Select EC2 service from the list of services
4. Create a security group by allowing SSH(Port: 22) and HTTP(Port: 80) from Network & Security section on the left panel of EC2.
5. Then back to EC2 and click Launch Instance.
6. Then create EC2 instances with mentioned AMIs.
 - a. Amazon Linux 2
 - b. Ubuntu
 - c. Red Hat Enterprise Linux

CONFIGURATION STEPS:

1. AMAZON LINUX
 - Name the instance and select OS Image



- Select AMI (free tier eligible for test purpose)

Launch an instance | EC2 | ap-south-1

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:

Dashboard | EC2 | u... Cokonet Academy mailtosinsha-Gmail All Bookmarks

aws IAM EC2

EC2 > Instances > Launch an instance

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-048f4445314bcaa09 (64-bit (x86), uefi-preferred) / ami-07efda20bfc979493 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs Free tier eligible

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.11.20260406.2 x86_64 HVM kernel-6.1

Architecture 64-bit (x86) **Boot mode** uefi-preferred **AMI ID** ami-048f4445314bcaa09 **Publish Date** 2026-04-06 **Username** ec2-user Verified provider

▼ Instance type Info | Get advice

Instance type t3.micro Family: t3 2 vCPU 1 GiB Memory Current generation: true Free tier eligible
On-Demand Linux base pricing: 0.0112 USD per Hour On-Demand SUSE base pricing: 0.0112 USD per Hour All generations

▼ Summary

Number of instances 1 Info

Software Image (AMI) Amazon Linux 2023 AMI 2023.11....read more
ami-048f4445314bcaa09

Virtual server type (instance type) t3.micro

Firewall (security group) webserver-sg

Storage (volumes) 1 volume(s) - 8 GiB

Cancel Launch instance Preview code

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Type here to search 7:16 PM 13-Apr-26

- Select the Keypair and security group already created

Launch an instance | EC2 | ap-south-1

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:

Dashboard | EC2 | u... Cokonet Academy mailtosinsha-Gmail All Bookmarks

aws IAM EC2

EC2 > Instances > Launch an instance

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required mumbai_keypair Create new key pair

▼ Network settings Info Edit

Network vpc-0492ac15bccee436d Info

Subnet No preference (Default subnet in any availability zone) Info

Auto-assign public IP Enable Info

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group Select existing security group

▼ Summary

Number of instances 1 Info

Software Image (AMI) Amazon Linux 2023 AMI 2023.11....read more
ami-048f4445314bcaa09

Virtual server type (instance type) t3.micro

Firewall (security group) webserver-sg

Storage (volumes) 1 volume(s) - 8 GiB

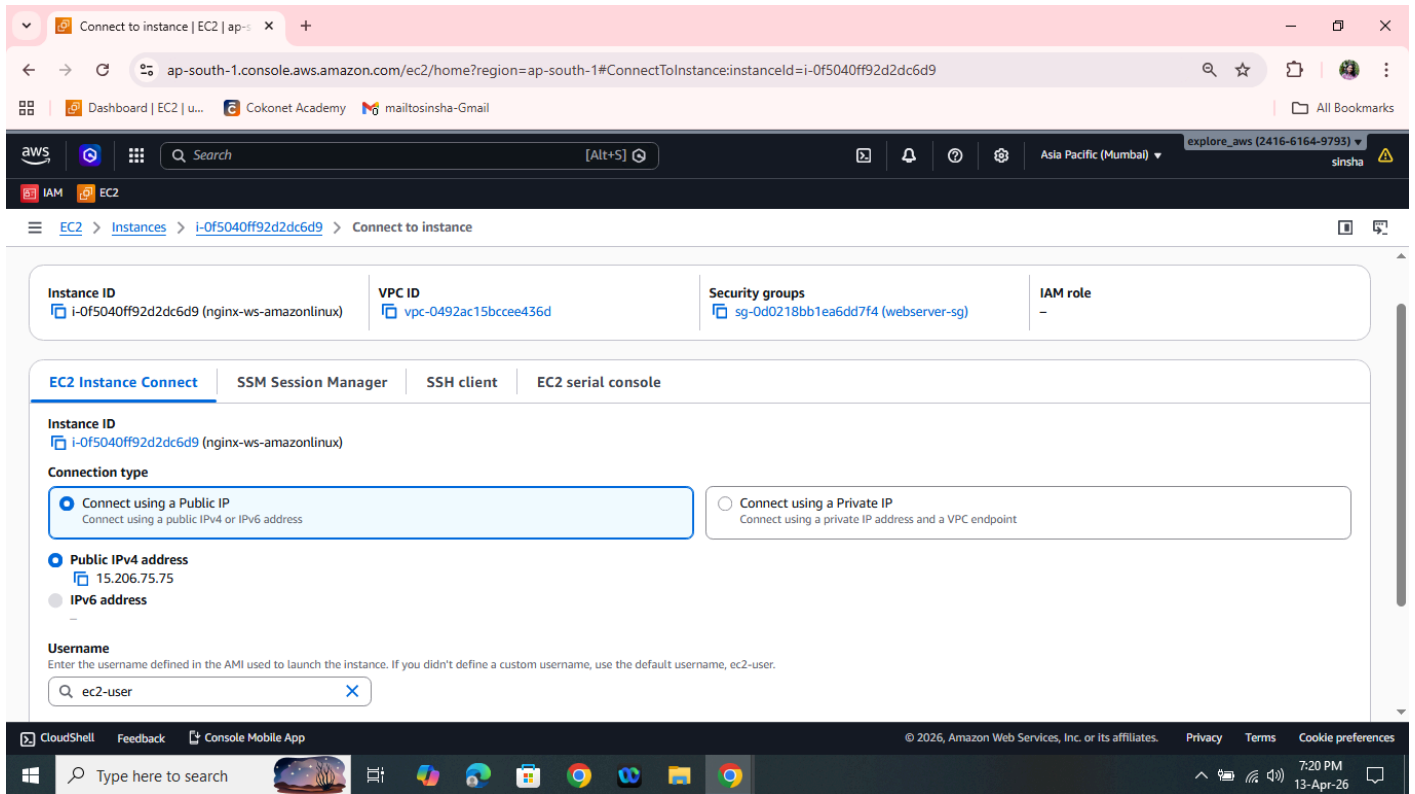
Cancel Launch instance Preview code

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Type here to search 7:18 PM 13-Apr-26

- Then click on Launch instance.

- After an instance has been created login into Amazon Linux instance by Amazon browser based login as shown below.
- In the case of Redhat use Mobaxterm or Putty for login to the terminal.



Do the following command execution on the coming terminal

1. `sudo su`
2. `dnf update -y`
3. `dnf install nginx -y`
4. `systemctl start nginx`

The screenshot shows the AWS CloudShell interface. The terminal output displays the installation and status of the nginx service. It shows that the service is loaded and active, with the configuration file /etc/nginx/nginx.conf being tested successfully. The service is running as a systemd unit, and the main PID is 25756. The tasks are 3 (limit: 1014), and the memory usage is 3.4M. The CPU usage is 57ms. The CGroup is /system.slice/nginx.service. The terminal also shows the execution of the systemctl start nginx command and the systemctl status nginx command. The status output shows that the service is active (running) since Mon 2026-04-13 16:26:06 UTC, 15s ago. The process is 25753 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS). The process is 25754 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS). The process is 25755 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS). The main PID is 25756 (nginx). The tasks are 3 (limit: 1014). The memory usage is 3.4M. The CPU usage is 57ms. The CGroup is /system.slice/nginx.service. The CGroup shows three processes: 25756 "nginx: master process /usr/sbin/nginx", 25757 "nginx: worker process", and 25758 "nginx: worker process". The terminal also shows the execution of the systemctl status nginx command and the systemctl start nginx command. The status output shows that the service is active (running) since Mon 2026-04-13 16:26:06 UTC, 15s ago. The process is 25753 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS). The process is 25754 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS). The process is 25755 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS). The main PID is 25756 (nginx). The tasks are 3 (limit: 1014). The memory usage is 3.4M. The CPU usage is 57ms. The CGroup is /system.slice/nginx.service. The CGroup shows three processes: 25756 "nginx: master process /usr/sbin/nginx", 25757 "nginx: worker process", and 25758 "nginx: worker process".

```
o nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
   Active: inactive (dead)
[root@ip-172-31-36-125 ec2-user]# systemctl start nginx
[root@ip-172-31-36-125 ec2-user]# systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
   Active: active (running) since Mon 2026-04-13 16:26:06 UTC; 15s ago
     Process: 25753 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 25754 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 25755 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
    Main PID: 25756 (nginx)
      Tasks: 3 (limit: 1014)
     Memory: 3.4M
        CPU: 57ms
    CGroup: /system.slice/nginx.service
            └─25756 "nginx: master process /usr/sbin/nginx"
              └─25757 "nginx: worker process"
                └─25758 "nginx: worker process"

Apr 13 16:26:06 ip-172-31-36-125.ap-south-1.compute.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Apr 13 16:26:06 ip-172-31-36-125.ap-south-1.compute.internal nginx[25754]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Apr 13 16:26:06 ip-172-31-36-125.ap-south-1.compute.internal nginx[25754]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Apr 13 16:26:06 ip-172-31-36-125.ap-south-1.compute.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[root@ip-172-31-36-125 ec2-user]#
```

i-0f5040ff92d2dc6d9 (nginx-ws-amazonlinux)

PublicIPs: 15.206.75.75 PrivateIPs: 172.31.36.125

- After nginx service is successfully running, copy the public IP of the instance and type in any browser as below:
 - `http://<IP ADDRESS OF THE INSTANCE>/`
 - Then the default page of nginx will be shown below.

The screenshot shows a web browser window with the address bar displaying "15.206.75.75". The page content is the default nginx welcome page, which includes the text "Welcome to nginx!", a message about successful installation, a link to the nginx.org website for documentation, and a thank you message.

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

- Do the same step for other mentioned os as well
- The execution command will be different for Ubuntu.

2. UBUNTU

The screenshot shows the AWS Management Console interface for launching an EC2 instance. The browser tabs include 'Launch an instance | EC2 | ap-southeast-1', 'EC2 Instance Connect | ap-southeast-1', and 'Welcome to nginx!'. The URL is 'ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:'. The console header shows the AWS logo, a search bar, and the region 'Asia Pacific (Mumbai)'. The main content area is titled 'Launch an instance' and features a grid of operating system logos: Amazon Linux, macOS, Ubuntu (selected), Windows, Red Hat, SUSE Linux, and Debian. Below the grid, the selected AMI is 'Ubuntu Server 24.04 LTS (HVM), SSD Volume Type' with AMI ID 'ami-05d2d839d4f73aafb'. The description states 'Ubuntu Server 24.04 LTS (HVM), EBS General Purpose (SSD) Volume Type. Support available from Canonical (http://www.ubuntu.com/cloud/services)'. The architecture is '64-bit (x86)'. The summary on the right indicates 1 instance, Canonical Ubuntu 24.04, t5.micro instance type, new security group, and 1 volume (8 GiB). The bottom of the console shows the Windows taskbar with the time 7:30 PM on 13-Apr-26.

Commands executed:

1. `sudo su`
2. `apt update`
3. `apt install nginx -y`

4. `systemctl status nginx`
5. Nginx will be running state by default
6. Start nginx cmd: `systemctl start nginx`

The screenshot displays the AWS CloudShell interface. The terminal window shows the output of the `systemctl status nginx` command, indicating that the `nginx.service` is loaded, active (running), and enabled. It also shows the process details for the master and worker processes. Below the terminal output, the instance details for `i-0e8dbb7577fcc29a2 (nginx-ws-ubuntu)` are visible, including public and private IP addresses.

Below the terminal window, a browser window is open, displaying the "Welcome to nginx!" page. The page contains the following text:

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.



3. REDHAT

The screenshot displays the AWS Management Console interface for launching an EC2 instance. The browser address bar shows the URL: `ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:`. The console header includes the AWS logo, a search bar, and navigation links for IAM and EC2. The main content area is titled "Launch an instance" and features a search bar with the text "nginx-ws-redhat".

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents | **Quick Start**

The "Quick Start" section displays a grid of OS images: Amazon Linux, macOS, Ubuntu, Windows, Red Hat, SUSE Linux, and Debian. The Red Hat image is highlighted.

Amazon Machine Image (AMI)

Red Hat Enterprise Linux 10 (HVM), SSD Volume Type
ami-03793655b06c6e29a (64-bit (x86)) / ami-0ad2259b693ba2cad (64-bit (Arm))
Free tier eligible

Summary

Number of instances | [Info](#)
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.11....[read more](#)
ami-048f4445314bcaa09

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

The bottom section of the console shows the "Amazon Machine Image (AMI)" details for the selected Red Hat image. It includes a description, architecture (64-bit (x86)), AMI ID (ami-03793655b06c6e29a), publish date (2026-02-18), and username (ec2-user). The image is marked as a "Verified provider".

Amazon Machine Image (AMI)

Red Hat Enterprise Linux 10 (HVM), SSD Volume Type
ami-03793655b06c6e29a (64-bit (x86)) / ami-0ad2259b693ba2cad (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs
Free tier eligible

Description
Red Hat Enterprise Linux version 10 (HVM), EBS General Purpose (SSD) Volume Type
Provided by Red Hat, Inc.

Architecture | **AMI ID** | **Publish Date** | **Username** | **Verified provider**

64-bit (x86) | ami-03793655b06c6e29a | 2026-02-18 | ec2-user | [Verified provider](#)

Instance type | [Info](#) | [Get advice](#)

Instance type

t3.micro
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0112 USD per Hour On-Demand SUSE base pricing: 0.0112 USD per Hour
On-Demand Windows base pricing: 0.0204 USD per Hour
Free tier eligible

[All generations](#) [Compare instance types](#)

The bottom of the console shows the "Instance type" section, which lists the selected instance type (t3.micro) and provides details about its family, vCPU, memory, and pricing. The instance type is marked as "Free tier eligible".

Launch an instance | EC2 | ap-south-1 | Instances | EC2 | ap-south-1

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#Instances:

Dashboard | EC2 | IAM | EC2

Search [Alt+S]

Asia Pacific (Mumbai) explore_aws (2416-6164-9793) sinsha

EC2 > Instances

Instances (1/3) Info

Saved filter sets: Choose filter set

Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
nginx-ws-redhat	i-01e011a589ab53b16	Running	t3.micro	Initializing	View alarms +	ap-south-1a
nginx-ws-ubuntu	i-0e8dbb7577fcc29a2	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1a
nginx-ws-amazonlinux	i-0f5040ff92d2dc6d9	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1a

i-01e011a589ab53b16 (nginx-ws-redhat)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ Instance summary info

Instance ID: i-01e011a589ab53b16

Public IPv4 address: 52.66.207.105 | open address

Private IPv4 addresses: 172.31.33.56

Public DNS: ec2-52-66-207-105.ap-south-1.compute.amazonaws.com

IPv6 address: -

Inst Copy public IPv4 address to clipboard

Running

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7:40 PM 13-Apr-26

MobaXterm

Terminal Sessions View X server Tools Games Settings Macros Help

Session Servers Tools Games

Quick connect...

User sessions

- 13.126.168.160 (ubuntu)
- 15.207.51.123 (ec2-user)
- 3.111.157.179 (ec2-user)
- 52.66.203.174 (ec2-user)
- 52.66.203.174 (ec2-user) (1)
- 52.66.207.105 (ec2-user)

Session settings

SSH Telnet Rsh Xdmcp RDP VNC FTP SFTP Serial File Shell Browser Mosh Aws S3 WSL

Basic SSH settings

Remote host: 52.66.207.105 Username: ec2-user Port: 22

Advanced SSH settings

Terminal settings Network settings Bookmark settings

Execute command: Remote environment: Interactive shell

SSH-browser type: SFTP protocol

Use private key: D:\COKONETAWS\mumbai_key

Execute macro at session start: <none>

OK Cancel

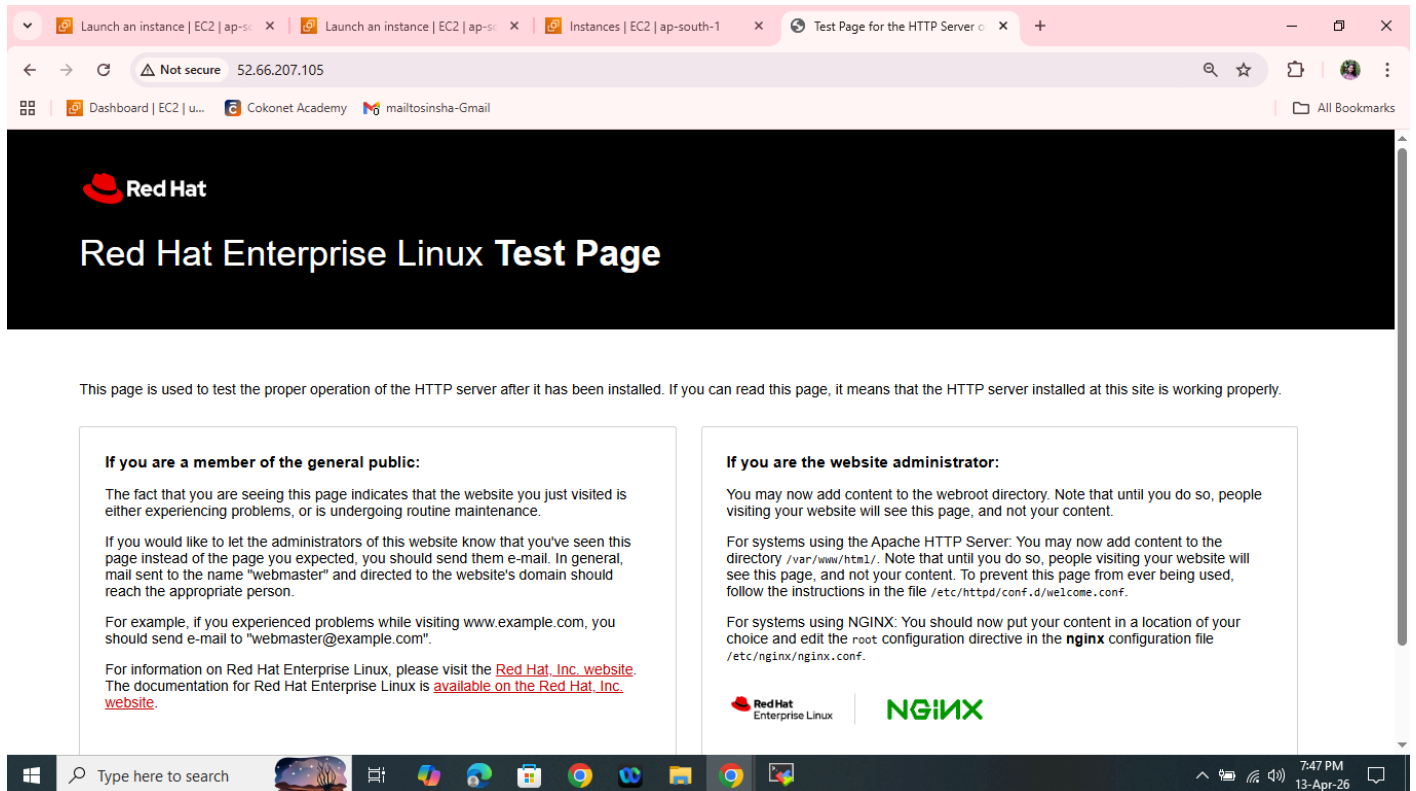
UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net

Type here to search

7:42 PM 13-Apr-26

Command executed:

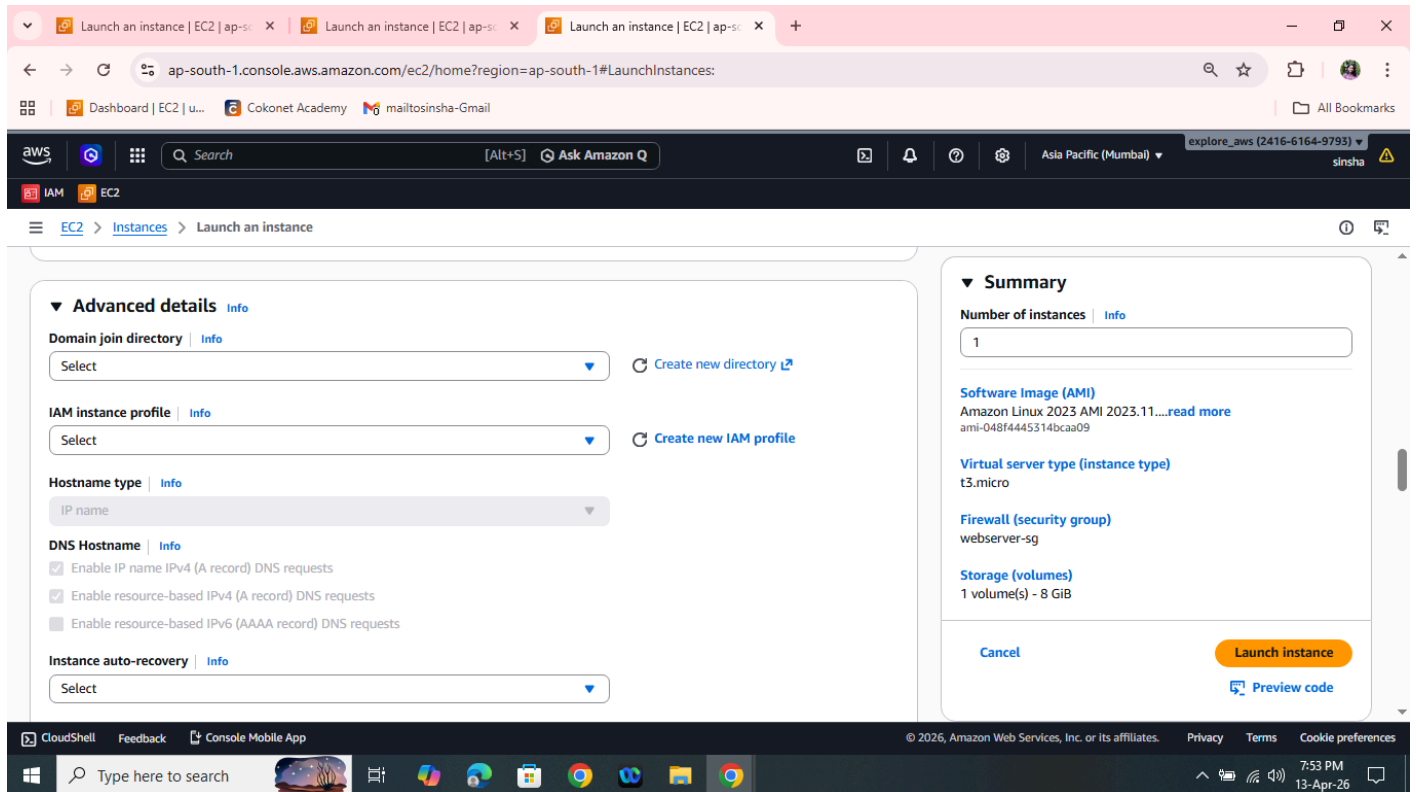
- 1 dnf update
- 2 dnf install nginx -y
- 3 systemctl status nginx
- 4 systemctl start nginx
- 5 systemctl status nginx

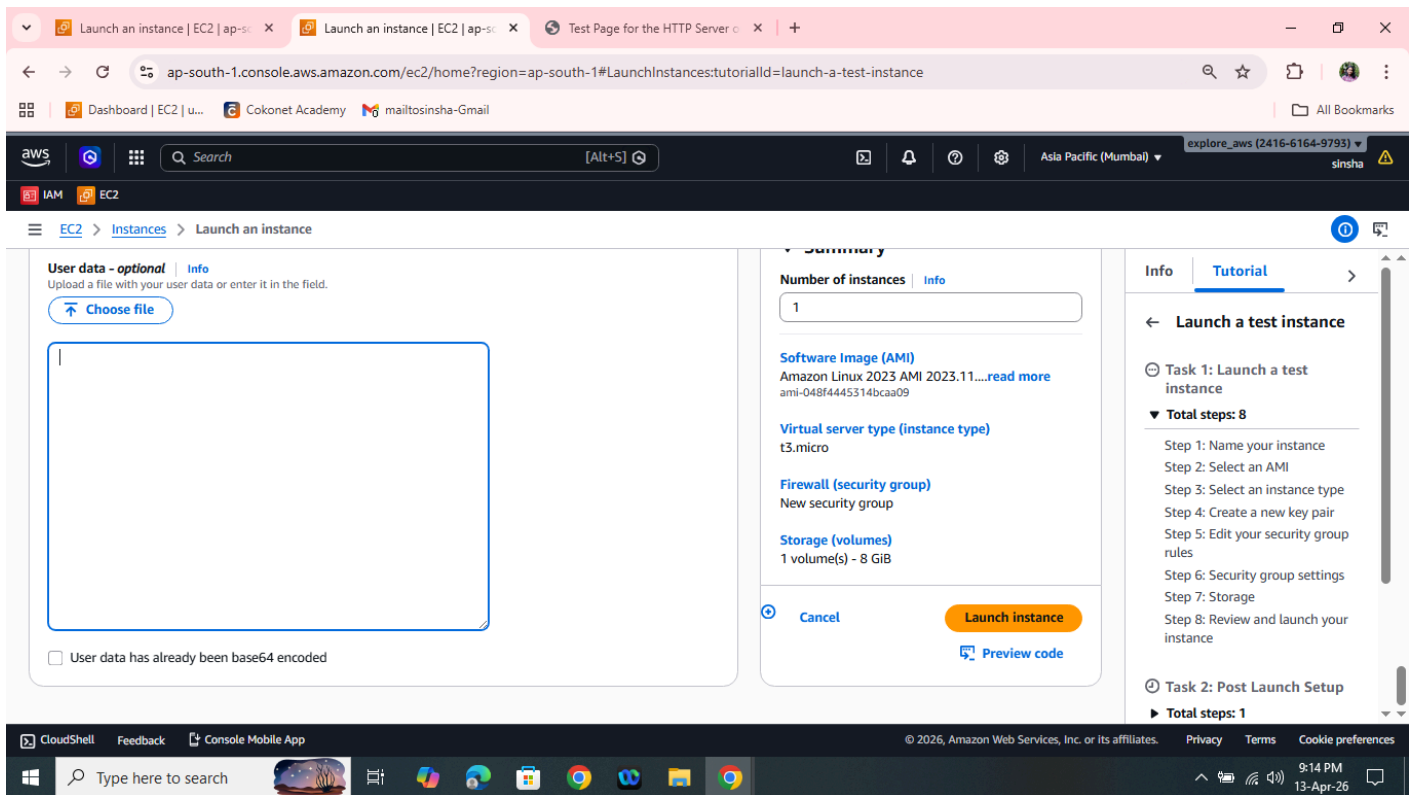


Configure the instance with Nginx using user data (a Bash script). In both cases, a default web page must be displayed.

CONFIGURATION STEPS:

1. Prerequisites will be the same as above.
2. Click on Launch instance from EC2 service from aws
3. Name the instance and select OS Image, instance type, Key pair and Security group
4. Then click on Advance details and come down where you see **User data**

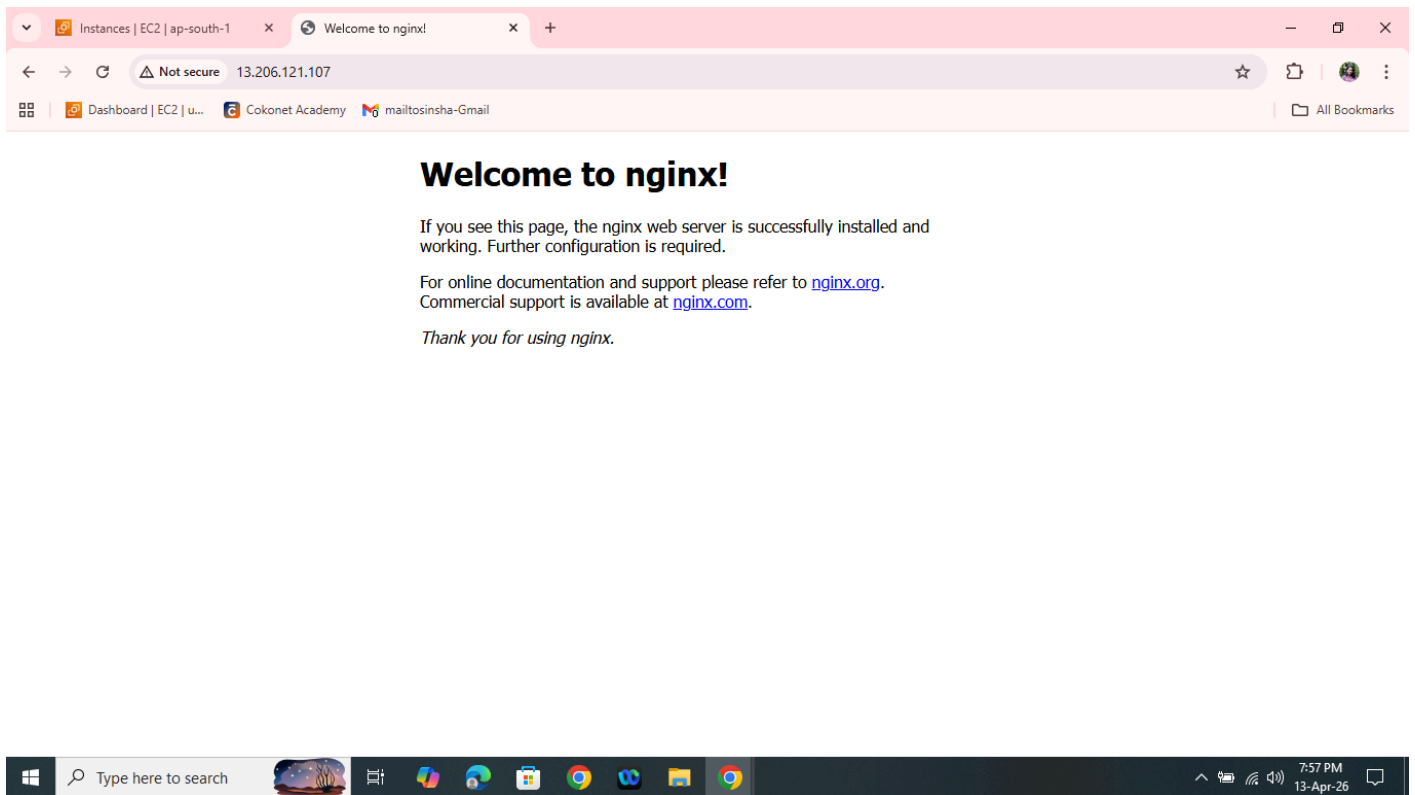




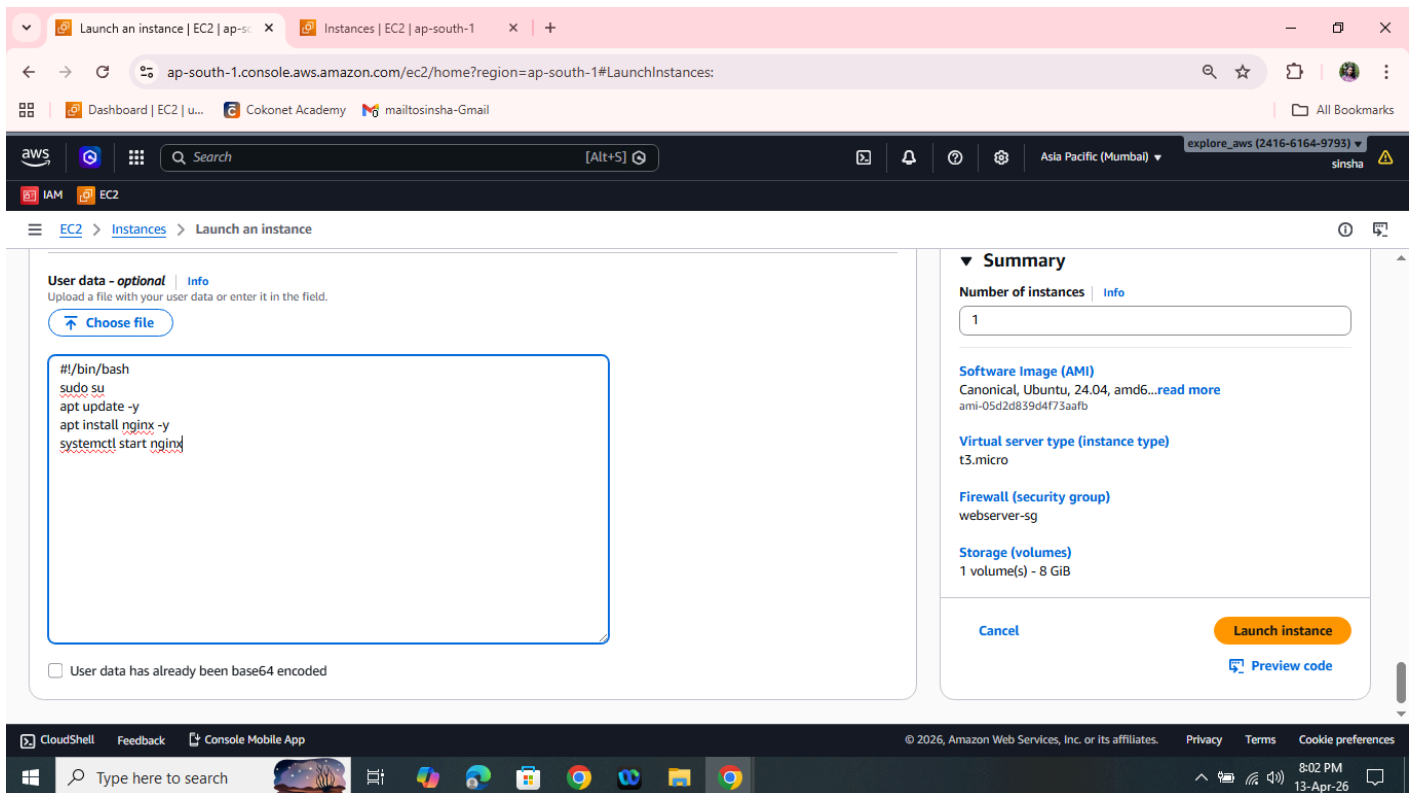
5. Fill the user data details according to the AMI you have selected, then Launch instance.
6. Once the instance is launched open the public ip the browser, where you can see the default page nginx

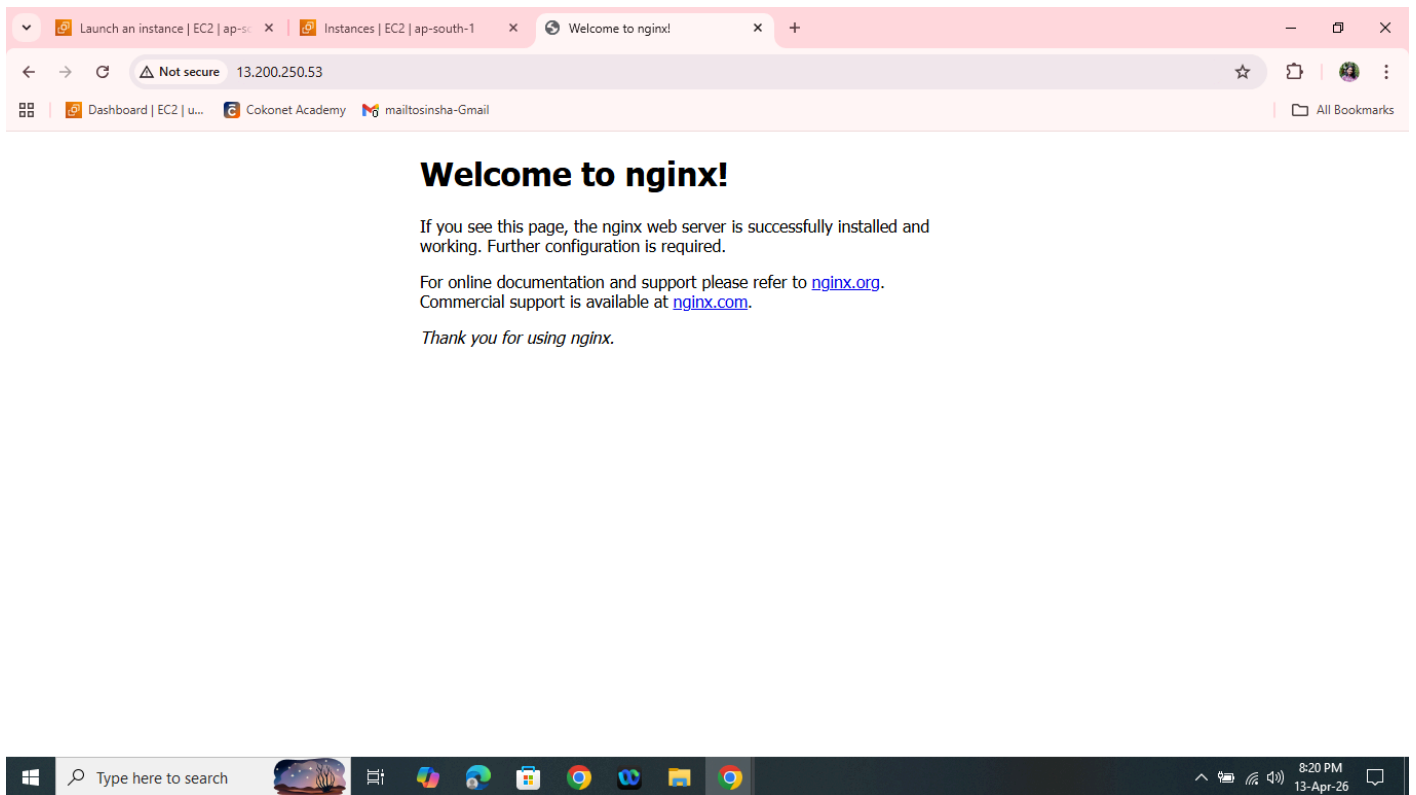
For AMAZON LINUX

```
#!/bin/bash
sudo su
dnf install nginx -y
systemctl start nginx
```

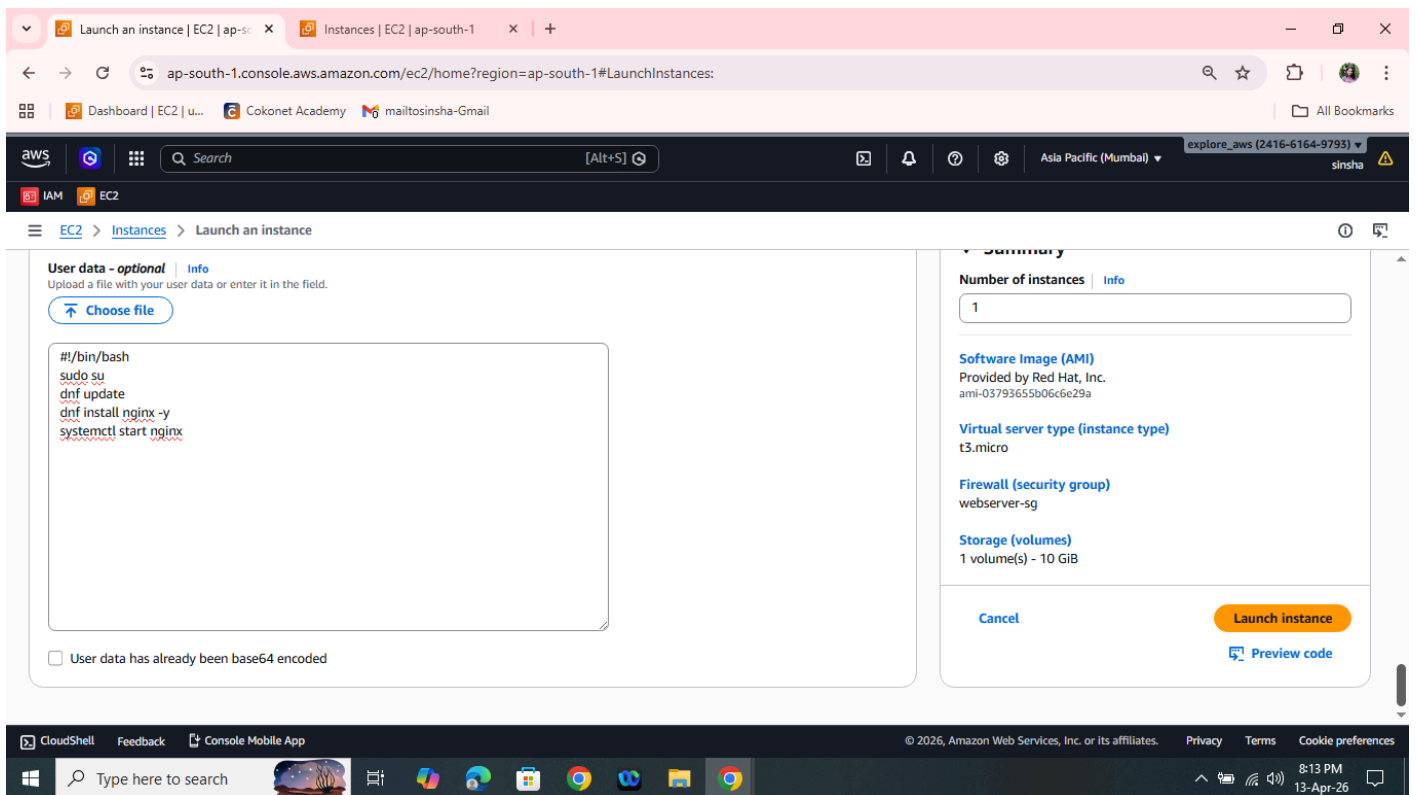


2. UBUNTU





3. REDHAT



Launch an instance | EC2 | ap-s...

Instances | EC2 | ap-south-1

Test Page for the HTTP Server

Not secure13.234.17.192

Dashboard | EC2 | u...Cokonet Academymailtosinsha-GmailAll Bookmarks



Red Hat Enterprise Linux Test Page

This page is used to test the proper operation of the HTTP server after it has been installed. If you can read this page, it means that the HTTP server installed at this site is working properly.

If you are a member of the general public:

The fact that you are seeing this page indicates that the website you just visited is either experiencing problems, or is undergoing routine maintenance.

If you would like to let the administrators of this website know that you've seen this page instead of the page you expected, you should send them e-mail. In general, mail sent to the name "webmaster" and directed to the website's domain should reach the appropriate person.

For example, if you experienced problems while visiting [www.example.com](#), you should send e-mail to "webmaster@example.com".

For information on Red Hat Enterprise Linux, please visit the [Red Hat, Inc. website](#). The documentation for Red Hat Enterprise Linux is [available on the Red Hat, Inc. website](#).

If you are the website administrator:

You may now add content to the webroot directory. Note that until you do so, people visiting your website will see this page, and not your content.

For systems using the Apache HTTP Server: You may now add content to the directory `/var/www/html/`. Note that until you do so, people visiting your website will see this page, and not your content. To prevent this page from ever being used, follow the instructions in the file `/etc/httpd/conf.d/welcome.conf`.

For systems using NGINX: You should now put your content in a location of your choice and edit the root configuration directive in the **nginx** configuration file `/etc/nginx/nginx.conf`.



Type here to search8:22 PM13-Apr-26